



Discrete Mathematics 2025 Spring



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- 9.1 Binary Operations and Their Properties
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■ 9.1.1 Binary and Unary Operations

- Definition of Binary Operations and Examples
- Definition of Unary Operations and Examples
- Notation of Operations

■ 9.1.2 Properties of Binary Operations

- Commutativity, Associativity, Idempotency, Distributivity, Absorption.
- Special Elements: Identity Element, Zero Element, Invertible Element.
- Cancellation Laws

Definition of Binary Operations and Examples

■ **Definition 9.1:** Let S be a set. A function $f : S \times S \rightarrow S$ is called a *binary operation* on S , or simply a binary operation. We also say that S is **closed** under f .

■ **Examples:**

(1) Binary operations on the set of natural numbers $\mathbf{N}$: addition and multiplication.

Note: Subtraction and division do not satisfy closure, as the result may not be a natural number.

(2) Binary operations on the set of integers $\mathbf{Z}$: addition, subtraction, and multiplication.

Note: Division may produce non-integer results, thus lacking closure.

(3) Binary operations on the set of nonzero real numbers $\mathbf{R}^*$: multiplication and division.

Note: Addition and subtraction do not satisfy closure, as results can include zero.

■ Examples:

(4) Let $S = \{a_1, a_2, \dots, a_n\}$, $a_i \circ a_j = a_i$, $\circ$ is a binary operation on S .

(5) Let $M_n(\mathbb{R})$ denote the set of all n ($n \geq 2$) real matrices, then

$$M_n(\mathbb{R}) = \left\{ \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \mid a_{ij} \in \mathbb{R}, i, j = 1, 2, \dots, n \right\}$$

Matrix addition and multiplication are both binary operations on $M_n(\mathbb{R})$.

(6) Binary operations on the power set $P(S)$: $\cup$ 、 $\cap$ 、 $-$ 、 $\oplus$.

(7) S^S be the set of all functions from S , the **composition operation** $\circ$ is a binary operation on S^S .

↳ Unary operations and n -ary operations

- **Definition 9.2:** Let S be a set, A function $f : S \rightarrow S$ is called a **unary operation** on S , or simply a unary operation.
 - More generally, a function $f : \underbrace{S \times S \times \dots \times S}_n \rightarrow S$ is called an **n -ary operation** on S , or simply an **n -operation**.
- **Examples:** The following are all unary operations:
 - (1) The **negation** (taking the additive inverse) operation on $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$.
 - (2) The **reciprocal** (taking the multiplicative inverse) operation on the set of nonzero rational numbers $\mathbb{Q}^*$ and the set of nonzero real numbers $\mathbb{R}^*$.
 - (3) The **complex conjugate** operation on the set of complex numbers $\mathbb{C}$.
 - (4) On the power set $P(S)$ with universe S , the **absolute complement** operation $\sim$.
 - (5) Let A be the set of all bijective functions on S $A \subseteq S^S$, the **inverse function** operation.
 - (6) On $M_n(\mathbb{R})$ ($n \geq 2$) the **matrix transpose** operation.

↳ Binary, Unary, and n-ary Operators and Representations

- **Operators:** $\circ$, $*$, $\cdot$, $\oplus$, $\otimes$ are used to denote binary or unary operations.
 - For a *binary operation* $\circ$, if x and y operate to produce z , it is denoted as $x \circ y = z$.
 - For a *unary operation* $\circ$, the result of applying it to x is written as $\circ x$.
 - For an *n-ary operation*, we write $\circ(a_1, a_2, \dots, a_n) = b$.
- Ways to represent binary or unary operations: *formulas*、*operation tables*.

Note: In the same problem or context, different operations should be represented using different operator symbols.

Formula and Table Representations of Operations

Formula Representation

- Example:** Let R be the set of real numbers. Define a binary operation R as: $\forall x, y \in R, x * y = x$.

Such as: $3 * 4 = 3, 0.5 * (-3) = 0.5$

- Tables Representation:** used to represent unary and binary operations on finite sets.

Operation Table for
Unary Operations

	$\circ a_i$
a_1	$\circ a_1$
a_2	$\circ a_2$
...	...
a_n	$\circ a_n$

Operation Table for Binary Operations

$\circ$	a_1	a_2	...	a_n
a_1	$a_1 \circ a_1$	$a_1 \circ a_2$		$a_1 \circ a_n$
a_2	$a_2 \circ a_1$	$a_2 \circ a_2$		$a_2 \circ a_n$
...				
a_n	$a_n \circ a_1$	$a_n \circ a_2$		$a_n \circ a_n$

9.1.1 Binary and Unary Operations

↳ Example of Operation Table Representations

- **Example:** Set $S = \{a, b\}$, Provide the operation tables $\oplus$ and $\sim$ on the power set $P(S)$.

$\oplus$ Operation Table

$\oplus$	$\emptyset$	$\{a\}$	$\{b\}$	$\{a, b\}$
$\emptyset$	$\emptyset$	$\{a\}$	$\{b\}$	$\{a, b\}$
$\{a\}$	$\{a\}$	$\emptyset$	$\{a, b\}$	$\{b\}$
$\{b\}$	$\{b\}$	$\{a, b\}$	$\emptyset$	$\{a\}$
$\{a, b\}$	$\{a, b\}$	$\{b\}$	$\{a\}$	$\emptyset$

$\sim$ Operation Table

X	$\sim X$
$\emptyset$	$\{a, b\}$
$\{a\}$	$\{b\}$
$\{b\}$	$\{a\}$
$\{a, b\}$	$\emptyset$

↳ Example of Operation Table Representation

- **Example:** Let $Z_5 = \{0, 1, 2, 3, 4\}$, $\oplus$, $\otimes$ represent addition and multiplication modulo 5, respectively.

$\oplus$ Operation Table

$\oplus$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\otimes$ Operation Table

$\otimes$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

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Binary Operation: Commutative, Associative, Idempotent Laws

- **Definition 9.3:** Let $\circ$ be a binary operation on S ,
 - (1) If for all $x, y \in S$, $x \circ y = y \circ x$, then the operation is said to satisfy the **commutative law** (or is **commutative**) on S .
 - (2) If for all $x, y, z \in S$, $(x \circ y) \circ z = x \circ (y \circ z)$, then the operation is said to satisfy the **associative law** (or is **associative**) on S .
 - (3) If for all $x \in S$, $x \circ x = x$, then the operation is said to satisfy the **idempotent law** (or is **idempotent**) on S .

- **Example:** $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ denote the sets of integers, rational numbers, and real numbers, respectively. $M_n(\mathbb{R})$ denotes the set of all $n \times n$ real matrices, $n \geq 2$, $P(B)$ power set, A^A set of all functions from A to A , $|A| \geq 2$.

Set	Operations	Commutative law	Associative law	Idempotent law
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}$	+	Yes	Yes	No
	$\times$	Yes	Yes	No
$M_n(\mathbb{R})$	+	Yes	Yes	No
	$\times$	No	Yes	No
$P(B)$	$\cup$	Yes	Yes	Yes
	$\cap$	Yes	Yes	Yes
	$-$	No	No	No
	$\oplus$	Yes	Yes	No
A^A	$\circ$	No	Yes	No

↳ Distributive & Absorption Laws (Two Binary Operations)

■ **Definition 9.4:** Let $\circ$ and $*$ be two distinct binary operations on S ,

(1) If for all $x, y, z \in S$

$$(x * y) \circ z = (x \circ z) * (y \circ z)$$

$$z \circ (x * y) = (z \circ x) * (z \circ y)$$

then we say that the operation $\circ$ *distributes* over the operation $*$.

(2) If both $\circ$ and $*$ are commutative, and for all $x, y \in S$

$$x \circ (x * y) = x$$

$$x * (x \circ y) = x$$

then we say that the operations $\circ$ and $*$ satisfy the *absorption law*.

- **Example:** $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ denote the sets of integers, rational numbers, and real numbers, respectively; $M_n(\mathbb{R})$ denotes the set of all $n \times n$ real matrices, $n \geq 2$, $P(B)$ power set, A^A set of all functions from A to A , $|A| \geq 2$.

Set	Operations	Distributive Law	Absorption Law
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}$	+ and $\times$	$\times$ to + distribute	No
		+ to $\times$ Not distribute	
$M_n(\mathbb{R})$	Matric + and matrix $\times$	$\times$ to + distribute	No
		+ to $\times$ Not distribute	
$P(B)$	$\cup$ and $\cap$	$\cup$ to $\cap$ distribute	Yes
		$\cap$ to $\cup$ distribute	
	$\cap$ and $\oplus$	$\cap$ to $\oplus$ distribute	No
		$\oplus$ to $\cap$ Not distribute	